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Title: Proposal for a new Technical Recommendation on the “technical requirement and implementation strategy for Global IMEI Registry” ITU-T Q.GIR

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Abstract: This contribution proposes the new Technical Recommendation ITU-T Q.GIR for the technical requirement and implementation strategy for Global IMEI Registry.

Annex A

A.1 justification for proposed draft new ITU-T Q.GIR "Technical requirement and implementation strategy for Global IMEI Registry"

Question:	Q15/11	Proposed new ITU-T Recommendation	Geneva, 10-19 May 2023
Reference and title:	ITU-T Q.GIR "Technical requirement and implementation strategy for Global IMEI Registry"		
Base text:	SG11-C0000		Timing: 2025-03
Editor(s):	1. Pushpendra Kumar Singh Ministry of Communication India 2. Biren Karmakar C-DOT, Ministry of Communication India 3. Arun Kumar Bediya C-DOT, Ministry of Communication India		Approval process: Agreement

Scope

This Recommendation proposes the requirement for implementation of Global IMEI Registry and suggests the strategy through which it could be implemented.

Summary

With the growing usage of information and communications technology (ICT) equipment in people's daily lives in recent years, unwelcome side effects related to the increase in the sale, circulation and use of counterfeit ICT devices in the market has also been increased.

As described in the Recommendation ITU-T Q.5050 (Framework for solutions to combat counterfeit ICT devices), this has resulted in adverse consequences for multiple stakeholders such as users, network operators, genuine device manufacturers, traders and governments, including decreased security protection and quality of service for users and revenue losses to a range of stakeholders.

Section 10.1 of the above recommendation pointed out to "Prohibit the use of invalid and non-genuine device identifiers".

As the issue of IMEI re-programming still a major concern in combating counterfeiting ICT devices, section 9.3 of ITU-T Q.5050 mentioned about Reliable unique identifier - genuine ICT devices are required to have unique and persistent identifiers that are secure, in the sense that they cannot be changed by unauthorized entities, are unique to each equipment and have been assigned by the authorized assigner. Which implies that in order to reduce IMEI re-programming, it is required to identify that the IMEI is re-programmed.

When IMEI is re-programmed, newly implanted IMEI will be either random i.e. invalid, non-genuine IMEI, duplicate/already used IMEI or IMEI which is valid but not manufactured by the OEM. Invalid or non-genuine IMEI could be detected by MNO EIR by checking the IMEI format and matching with TAC information allocated by GSMA. Even already used IMEI can also be checked through CEIR. But to find out the IMEI which has not been manufactured by OEM/Manufacturers can be done only through the IMEI registry. The IMEI registry will contain the list of IMEIs which have been manufactured by the mobile manufacturers and issued for sale in the market/export. Some of the nations may already have implemented an IMEI registry for their country, but the issue remains the same when mobile devices are imported from other countries. This identifies the requirement of the Global IMEI registry. Participating nations will share IMEI produced in their country – which will be used by all other countries to find counterfeit mobile devices. ITU-T Q.5050 also mentioned about the Certification of the ICT device in section 10.2 to assist all stakeholders and guarantee the authenticity of the products, one possible solution should be to deploy a device lifecycle management system, based on these unique identifiers, that are able to track the ICT devices from the beginning of the manufacture process (including component origin, transportation and retail store where it will be sold) until it is delivered to the end-user - which could easily be implemented through Global IMEI Registry.

ITU-T Q.5052 recommendation urges the requirement of Duplicate IMEI detection across countries in section 7.5. However, this is related to the IMEI which is already in use by the subscribers. It does not include the IMEI which has been manufactured but not used in the network. Resolution 96 resolved that "ITU-T should consider solutions to be used to differentiate between authentic/genuine and counterfeit or tampered telecommunication/ICT devices, e.g. establishing a centralized national reference database of authorized equipment; Resolution 97 resolved that "ITU-T should, in collaboration with the relevant standards organizations, develop solutions to address the problem of duplication of unique identifiers". Global IMEI Registry will mostly resolve the problems such as IMEI cloning/duplication due to re-programming of IMEI.
Relations to ITU-T Recommendations or to other standards (approved or under development): ITU-T Resolution 96 - (Hammamet, 2016), "ITU Telecommunication Standardization Sector studies for combating counterfeit telecommunication/information and communication technology devices" ITU-T Resolution 97 - (Rev. Geneva, 2022), "Combating mobile telecommunication device theft" ITU-T Q.5050, "Framework for solutions to combat counterfeit ICT devices" ITU-T Q.5052, "Addressing mobile devices with a duplicate unique identifier" ITU-T QTR-RLB-IMEI, "Reliability of International Mobile station Equipment Identity (IMEI)"
Liaisons with other study groups or with other standards bodies: SG17, GSMA
Supporting members that are committing to contributing actively to the work item: C-DOT, India,

Annex B

Summary

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As described in the Recommendation ITU-T Q.5050 (Framework for solutions to combat counterfeit ICT devices), this has resulted in adverse consequences for multiple stakeholders such as users, network operators, genuine device manufacturers, traders and governments, including decreased security protection and quality of service for users and revenue losses to a range of stakeholders.

Section 10.1 of the above recommendation pointed out to "Prohibit the use of invalid and non-genuine device identifiers".

As the issue of IMEI re-programming still a major concern in combating counterfeiting ICT devices, section 9.3 of ITU-T Q.5050 mentioned about Reliable unique identifier - genuine ICT devices are required to have unique and persistent identifiers that are secure, in the sense that they cannot be changed by unauthorized entities, are unique to each equipment and have been assigned by the authorized assigner. Which implies that in order to reduce IMEI re-programming, it is required to identify that the IMEI is re-programmed.

When IMEI is re-programmed, newly implanted IMEI will be either random i.e. invalid, non-genuine IMEI, duplicate/already used IMEI or IMEI which is valid but not manufactured by the OEM. Invalid or non-genuine IMEI could be detected by MNO EIR by checking the IMEI format and matching with TAC information allocated by GSMA. Even already used IMEI can also be checked through CEIR. But to find out the IMEI which has not been manufactured by OEM/Manufacturers can be done only through the IMEI registry. The IMEI registry will contain the list of IMEIs which have been manufactured by the mobile manufacturers and issued for sale in the market/export. Some of the nations may already have implemented an IMEI registry for their country, but the issue remains the same when mobile devices are imported from other countries. This identifies the requirement of the Global IMEI registry. Participating nations will share IMEI produced in their country – which will be used by all other countries to find counterfeit mobile devices. ITU-T Q.5050 also mentioned about the Certification of the ICT device in section 10.2 to assist all stakeholders and guarantee the authenticity of the products, one possible solution should be to deploy a device lifecycle management system, based on these unique identifiers, that are able to track the ICT devices from the beginning of the manufacture process (including component origin, transportation and retail store where it will be sold) until it is delivered to the end-user - which could easily be implemented through Global IMEI Registry.

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Resolution 97 resolved that "ITU-T should, in collaboration with the relevant standards organizations, develop solutions to address the problem of duplication of unique identifiers".

Global IMEI Registry will mostly resolve the problems such as IMEI cloning/duplication due to re-programming of IMEI.

History

Edition	Recommendation	Approval	Study Group	Unique ID ^{1*}
1.0	ITU-T Q.GIR	---	11	--

Keywords

GIR, Global IMEI Registry, Combat counterfeit ICT devices, unique identifiers, replicated devices

¹ The number of the study group report.

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As of the date of approval of this Recommendation, ITU had not received notice of intellectual property, protected by patents, which may be required to implement this Recommendation. However, implementers are cautioned that this may not represent the latest information and are therefore strongly urged to consult the TSB patent database at <http://www.itu.int/ITU-T/ipr/>.

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Recommendation ITU-T Q.GIR

"Technical requirement and implementation strategy for Global IMEI Registry"

1 Scope

This Recommendation proposes the requirement for implementation of Global IMEI Registry and suggests strategy through which it could be implemented.

2 References

This ITU-T Recommendation and other references contain provisions which, through reference in this text, constitute provisions of this Recommendation. At the time of publication, the editions indicated were valid. All Recommendations and other references are subject to revision; users of this Recommendation are therefore encouraged to investigate the possibility of applying the most recent edition of the Recommendations and other references listed below. A list of the currently valid ITU-T Recommendations is regularly published. The reference to a document within this Recommendation does not give it, as a stand-alone document, the status of a Recommendation.

ITU-T Q.5050, "Framework for solutions to combat counterfeit ICT devices"

ITU-T Q.5052, "Addressing mobile devices with a duplicate unique identifier"

ITU-T QTR-RLB-IMEI, "Reliability of International Mobile station Equipment Identity (IMEI)"

ITU-T Q.CEIR, "Technical requirement, interfaces and generic functions of CEIR"

3 Definitions

3.1 Terms defined elsewhere

This Recommendation uses the following terms defined elsewhere:

counterfeit ICT device [ITU-T Q.5050]: An information and communication technology (ICT) device that explicitly infringes the trademark, copies hardware or software designs, or infringes brand or packaging rights of an original or authentic product and, in general, infringes applicable national and/or international technical standards, regulatory requirements or conformity processes, manufacturing licensing agreements, or other applicable legal requirements.

unique identifier [ITU-T Q.5050]: An identifier associated with a single device that aims to uniquely identify it.

3.2 Terms defined in this Recommendation

This Recommendation defines the following terms:

None.

4 Abbreviations and acronyms

This Recommendation uses the following abbreviations and acronyms:

API	Application Programming Interface
CEIR	Central Equipment Identity Register
EIR	Equipment Identity Register
GIR	Global IMEI Registry
GSMA	Global System for Mobile Communications (GSM) Association
ICT	Information and Communications Technology
IMEI	International Mobile Equipment Identity
ITU	International Telecommunication Union

LEA	Law Enforcement Agency
MNO	Mobile Network Operator
MSISDN	Mobile Station Integrated Services Digital Network
NIR	National IMEI Registry
ODM	Original Design Manufacturer
OEM	Original Equipment Manufacturers
REST	Representational State Transfer
RIR	Regional IMEI Registry
TAC	Type Allocation Code

5 Conventions

This Recommendation applies the following verbal forms for the expression of provisions:

- a) The keyword "is required to" indicate a requirement which must be strictly followed and from which no deviation is permitted, if conformance to this Recommendation is to be claimed.
- b) The keyword "should" indicate a requirement which is recommended but which is not absolutely required. Thus, this requirement need not be present to claim conformance.
- c) The keyword "may" indicate an optional requirement which is permissible, without implying any sense of being recommended. This term is not intended to imply that the vendor's implementation must provide the option, and the feature can be optionally enabled by the network operator/service provider. Rather, it means the vendor may optionally provide the feature and still claim conformance with this Recommendation.

6 General aspects

With the growing usage of information and communications technology (ICT) equipment in people's daily lives in recent years, unwelcome side effects related to the increase in the sale, circulation and use of counterfeit ICT devices in the market has also been increased.

As described in the Recommendation ITU-T Q.5050, counterfeit ICT device has resulted in adverse consequences for multiple stakeholders such as users, network operators, genuine device manufacturers, traders and governments, including decreased security protection and quality of service for users and revenue losses to a range of stakeholders.

Section 10.1 of the above recommendation pointed out to "Prohibit the use of invalid and non-genuine device identifiers".

In Section 9.3 of ITU-T Q.5050, it is recommended that authentic ICT devices must have unique and secure identifiers that are specific to each device and assigned by authorized assigners to combat the issue of IMEI re-programming and counterfeiting of ICT devices. It is important to have a mechanism in place for detecting IMEI counterfeiting in order to decrease the occurrence of IMEI re-programming.

When an IMEI is reprogrammed, the new IMEI could be arbitrary, incorrect, non-genuine, duplicate, recently used, or even legitimated but not manufactured by the original equipment manufacturer (OEM). Mobile network operator equipment identity register (MNO EIR) and central equipment identity register (CEIR) can detect fake or invalid IMEIs by comparing their format with the TAC issued by GSMA. However, only the IMEI registry can identify IMEIs that were not manufactured by OEMs or manufacturers.

The IMEI registry contains a list of IMEIs that are authorized for retail sale or export and manufactured by mobile device manufacturers. However, the problem still exists when mobile devices are imported from other countries. To solve this problem, a Global IMEI Registry is needed, where participating countries can exchange their IMEIs, which can be used by all countries to identify counterfeited mobile devices. One potential solution could be a device lifecycle management system that uses these unique identifiers to track ICT devices from the beginning of the manufacturing process (including component origin, shipping, and retail outlet) to when they are delivered to the end consumer. This is necessary because ITU-T Q.5050 also discusses the Certification of the ICT device in Section 10.2, which aims to support all stakeholders and ensure the authenticity of mobile devices

Section 7.5 of ITU-T Q.5052 urged for the detection of duplicate IMEI numbers across different countries, but this currently only applies to IMEIs that are already in use by subscribers and does not address the issue of unused IMEIs that have been produced. The Global IMEI Registry is needed to address this issue and it has

the potential to prevent the problems due to IMEI cloning and duplication that arise from re-programming of IMEIs.

7 Description of the System

Global IMEI Registry is a system to store and manage all the legitimate IMEIs manufactured by the mobile device manufacturers/OEMs/ODMs. Mobile devices available in GIR may be in-use by subscribers or kept in the stores for sale. It will verify as there are different mobile manufacturers in different countries, manufactured IMEIs to be collected from the local manufacturers of the country to prepare National IMEI Registry (NIR) – which needs to be shared with GIR to maintain IMEI data globally. GIR data could be used by any nation to check/validate the IMEI. As the mobile manufactured in one country could be exported to another country for sale and even mobile sold in one country could be used in another country. GIR data will be very useful for IMEI verification in those cases. GIR data can be utilized by the Government agencies, national telecom authority/ regulator, and law enforcement agencies. GIR data could be useful for the general public that can be accessible through mobile apps, web portals etc with required limited access.

8. Architecture

Following diagram shows National IMEI Registry architecture -

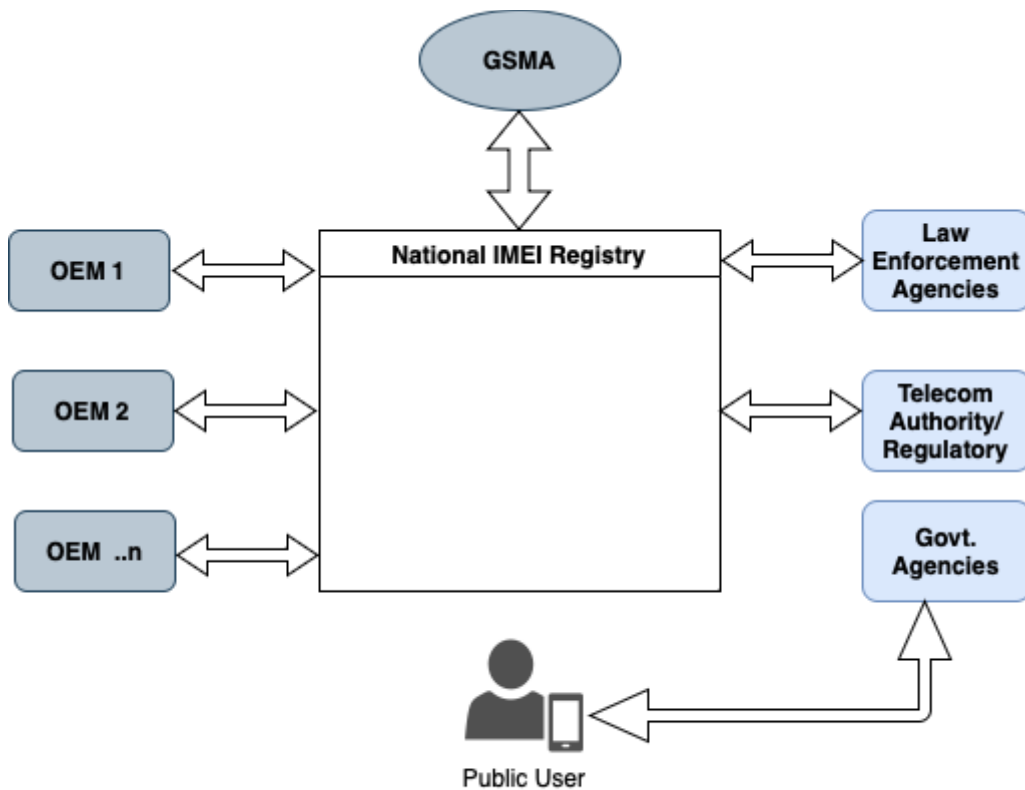


Figure 1: NIR Architecture

Multiple such National IMEI Registry will form Global IMEI Registry. If required, the Regional IMEI Registry (RIR) could also be prepared to connect NIR with GIR.

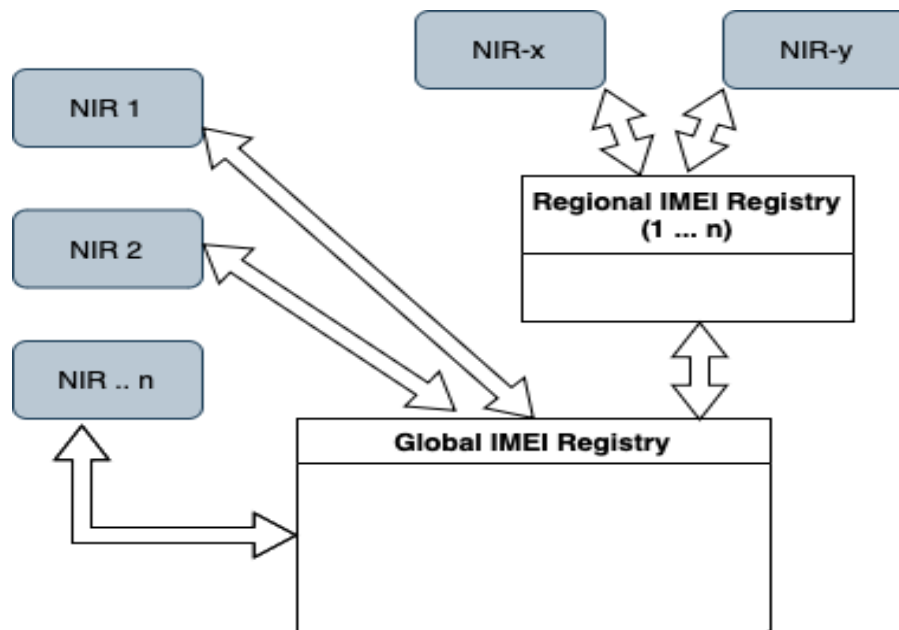


Figure 2: Multi-level GIR Architecture

9 External Interfaces

GIR requires interface only with NIR/RIR for IMEI data synchronization. Optionally it can interface with CEIR system.

All other national level stakeholders will be interfacing with NIR for IMEI verification. If required data is not available within the NIR, it will query GIR directly or through RIR to get the status of the requested IMEI preferably through REST based web-service, accordingly the result will be shown to the user.

10 System features

GIR will collect legitimate IMEI details from the manufacturers and provide details of the legitimate IMEIs including brand name, model, marketing name, manufacturing date. Option will also be there to delete IMEIs of the mobile devices which are scrapped by competent authority. IMEI insertion and deletion statistics will be available to the privileged users. There can be functionality to create different types of users with different privileges to access the system. Some other features such as report generation, activities performed by different users, list of imported IMEI from one nation to another, nation wise statistics, region wise statistics and global statistics etc. can be implemented based on requirement.

An Interface to communicate to NIR, RIR and GIR with CEIR can also be implemented.

11 Stakeholders

- 1) The mobile device original equipment manufacturer (OEM)/ original design manufacturer (ODM)/ Importers/ Brand owners.
- 2) Customs department
- 3) The Mobile Network Operators (MNOs).
- 4) The Law Enforcement Agencies and other Government agencies.
- 5) The Regulatory organisation

12 Non-functional Requirements

12.1 Performance Requirements

All other performance related to storage, memory, and processing should follow industry recommended practices to ensure resource requirements are minimised.

12.2 Reliability Requirements

In accordance with industry recommended practices, the system should undergo feature testing, load testing, and regression testing prior to release and/or deployment.

12.3 Availability Requirements

Reasonable efforts should be made to ensure the software system is available with an preferable uptime of 99.99%. The uptime is calculated as the percentage of time during the year in which the software system was available to the public.

12.4 Security Requirements

The system defined in this document must follow industry recommended practices for secure software development. At a minimum, the software development must practice the principle of least privilege for defining access-level requirements of the software system and its associated services. The production-release version of the software system must pass an automated dynamic application security testing/audit tool.
